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Agricultural baler system with bale stall identification

Abstract

An agricultural baler system includes a baler frame; a bale chamber; and a control system operatively coupled with the baler frame. The control system includes at least one first sensor configured for detecting at least one first operative condition associated with a baling operation and outputting at least one first operative condition signal corresponding to the at least one first operative condition. The control system further includes a controller system operatively coupled with the at least one first sensor. The control system is configured for receiving the at least one first operative condition signal and determining at least one operative parameter based at least in part on the at least one first operative condition signal. The at least one first operative parameter is associated with a bale of a crop material one of stalling and being about to stall in the bale chamber of a baler.

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention pertains to an agricultural baler system, and, more specifically, to an agricultural control system.

BACKGROUND OF THE INVENTION

(2) Agricultural harvesting machines, such as agricultural balers (which can be referred to balers), have been used to consolidate and package crop material (which, depending upon the application,

can also be referred to as forage, forage material, or forage crop material) so as to facilitate the storage and handling of the crop material for later use. Often, a mower-conditioner cuts and conditions the crop material for swath or windrow drying in the sun. When the cut crop material is properly dried (depending upon the application), an agricultural harvesting machine, such as an agricultural baler, which can be a round baler or a square baler, travels along the swaths or windrows (hereinafter, collectively referred to as windrows, unless otherwise specified) to pick up the crop material. In the case of round balers, the crop material is formed into cylindrically-shaped round bales, and in the case of square balers, the crop material is formed into small or large square bales. More specifically, pickups of the baler gather the cut and windrowed crop material from the ground, and then convey the cut crop material into a bale-forming chamber (which can be referred to as a bale chamber) within the baler. A drive mechanism operates to activate any pickups, augers, and/or a rotor of a feed mechanism. For a round baler, for instance, a conventional baling chamber may include a pair of opposing sidewalls with a series of rolls (which can be referred to as rollers) and belts that rotate and compress the crop material into a cylindrical shape. When the bale has reached a desired size and density, a wrapping assembly, which includes wrap material, may wrap the bale to ensure, at least in part, that the bale maintains its shape and density. The wrap material can include a film (such as a flexible plastic wrap) or a net (which can be referred to as net wrap). For example, wrap material may be used to wrap the bale of crop material. A cutting or severing mechanism of the wrapping assembly may be used to cut the wrap material once the bale has been wrapped. The wrapped bale may be ejected from the baler and onto the ground by, for example, raising a tailgate of the baler. The tailgate is then closed, and the cycle repeated as necessary and desired to manage the field of cut crop material.

(3) During baling, crop material and conditions are encountered in which the bale slows in rotation or stalls (stops rotating) within the bale chamber. This can occur when, for example, the bale chamber receives too much crop material and/or particularly heavy crop material (such as wet crop material). Driving the baler too hard or fast can cause the bale chamber to receive too much crop material. Further, occasionally, the bale chamber may suddenly receive a large quantity of crop material, such as a clump of crop material. The baler may not be able to handle these situations adequately, causing the rotation of the crop material within the bale chamber to slow or altogether stall. Further, the tendency of a bale to stall in the bale chamber may increase in conditions in which belts of the bale chamber are more prone to slip. For example, when baling in dusty or wet environments or baling with out-of-round bales (bales forming or formed in the bale chamber which are not cylindrical), power take-off (PTO) slippage and bale stalling are often observed.

(4) What is needed in the art is a way to detect when a bale of crop material has stalled or is about to stall in the bale chamber.

SUMMARY OF THE INVENTION

(5) The present invention provides an agricultural baler system which includes a control system which includes at least one sensor for detecting an operative condition associated with bale rotation speed, at least one sensor for detecting an operative condition associated with a reference speed, and determining whether the bale has stalled or is about to stall in light of what is sensed.

(6) The invention in one form is directed to a control system of an agricultural baler system, the agricultural baler system including a baler frame and a bale chamber at least one of coupled with and formed by the baler frame, the control system being operatively coupled with the baler frame, the control system including: at least one first sensor configured for: detecting at least one first operative condition associated with a baling operation; outputting at least one first operative condition signal corresponding to the at least one first operative condition; a controller system operatively coupled with the at least one first sensor and configured for: receiving the at least one first operative condition signal; determining at least one operative parameter based at least in part on the at least one first operative condition signal, the at least one first operative parameter being associated with a bale of a crop material one of stalling and being about to stall in a bale chamber

of a baler.
(7) The invention in another form is directed to an agricultural baler system including: a baler frame; a bale chamber at least one of coupled with and formed by the baler frame; a control system operatively coupled with the baler frame, the control system including: at least one first sensor configured for: detecting at least one first operative condition associated with a baling operation; outputting at least one first operative condition signal corresponding to the at least one first operative condition; a controller system operatively coupled with the at least one first sensor and configured for: receiving the at least one first operative condition signal; determining at least one operative parameter based at least in part on the at least one first operative condition signal, the at least one first operative parameter being associated with a bale of a crop material one of stalling and being about to stall in a bale chamber of a baler.
(8) The invention in yet another form is directed to a method of using an agricultural baler system, the method including the steps of: providing a baler frame, a bale chamber at least one of coupled with and formed by the baler frame, and a control system operatively coupled with the baler frame, the control system including at least one first sensor and a controller system operatively coupled with the at least one first sensor; detecting, by the at least one first sensor, at least one first operative condition associated with a baling operation; outputting, by the at least one first sensor, at least one first operative condition signal corresponding to the at least one first operative condition; receiving, by the controller system, the at least one first operative condition signal; and determining, by the controller system, at least one operative parameter based at least in part on the at least one first operative condition signal, the at least one first operative parameter being associated with a bale of a crop material one of stalling and being about to stall in a bale chamber of a baler.
(9) An advantage of the present invention is that it provides a way for ascertaining whether a bale has stalled or is about to stall in the bale chamber, and informing an operator.
(10) Another advantage of the present invention is that it provides a way for ascertaining whether a bale has stalled or is about to stall in the bale chamber, and automatically making at least one adjustment so as to prevent the stalling from occurring or undoing the stalling.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) For the purpose of illustration, there are shown in the drawings certain embodiments of the present invention. It should be understood, however, that the invention is not limited to the precise arrangements, dimensions, and instruments shown. Like numerals indicate like elements throughout the drawings. In the drawings:
(2) FIG. **1** illustrates schematically a side view of an exemplary embodiment of an agricultural baler system including an agricultural vehicle, formed as a tractor, and an agricultural baler, the agricultural baler system including a control system, in accordance with an exemplary embodiment of the present invention;
(3) FIG. **2** illustrates schematically a side cutaway view of the internal workings of the agricultural baler of FIG. **1**, in accordance with an exemplary embodiment of the present invention; and
(4) FIG. **3** illustrates a flow diagram showing a method of using the agricultural baler system, in accordance with an exemplary embodiment of the present invention.
DETAILED DESCRIPTION OF THE INVENTION
(5) The terms “forward”, “rearward”, “left” and “right”, when used in connection with the agricultural vehicle, agricultural baler, and/or components thereof are usually determined with reference to the direction of forward operative travel of the agricultural vehicle and/or agricultural baler, but they should not be construed as limiting. The terms “longitudinal” and “transverse” are

determined with reference to the fore-and-aft direction of the agricultural vehicle and/or agricultural baler and are equally not to be construed as limiting. The terms “downstream” and “upstream” are determined with reference to the intended direction of crop material flow during operation, with “downstream” being analogous to “rearward” and “upstream” being analogous to “forward.”

(6) Referring now to the drawings, and more particularly to FIG. **1**, there is shown an embodiment of an agricultural system **135** (which can be referred to as an agricultural machine system **135**) including an agricultural work vehicle **100** (which can be referred to as a work vehicle, or an agricultural vehicle) and an agricultural machine **101** (which can be referred to as an agricultural implement, or an implement), which is being towed by, and thus coupled with, agricultural work vehicle **100**, in accordance with the present invention, to perform an agricultural operation within a field, namely, baling. As shown, work vehicle **100** can be configured as an agricultural tractor, and agricultural machine **101** can be configured as a baler **101**, such as a round baler **101**, in accordance with an exemplary embodiment of the present invention, tractor **100** pulling baler **101** to perform a baling operation within the field (in this case, agricultural machine system **135** is an agricultural baler system **135**). Further, agricultural machine system **135** includes a control system **129** (which can be referred to as a unified control system **129**). Unified control system **129** includes control system **114** of tractor **100**, control system **122** of baler **101**, and, optionally, a control system (not shown) of a data center (not shown) that is cloud-based, Internet-based, and/or remotely located (this control system of the data center can be substantially similar to control systems **114**, **122**, having a controller, a processor, memory, data, and instructions, as described below with respect to control systems **114**, **122**). Control system **114** includes controller **115** (which can be a universal control module), and control system **122** includes controller **123**. Further, unified system **129** can be said to include controller system **128**, which includes controllers **115**, **123**. Control system **129**, and thus also control systems **114**, **122**, are operatively coupled with each of frames **104**, **130**. Control system **114**, in whole or in part, is further included as part of work vehicle **100**, and control system **122**, in whole or in part, is further included as part of baler **101**.

(7) Work vehicle **100** can be an operator-driven tractor or an autonomous tractor. However, in some embodiments, work vehicle **100** may correspond to any other suitable vehicle configured to tow a baler across a field or that is otherwise configured to facilitate the performance of a baling operation, including an autonomous baling vehicle. Additionally, as shown, baler **101** can configured as a round baler configured to generate round bales (alternatively, though not shown, the baler can be a square baler, configured to generate small or large square bales). It should be further appreciated that baler **101**, while shown as being towed by tractor **100**, may also be a self-propelled baler that does not rely on a separate vehicle for propulsion and/or power to function.

(8) Work vehicle **100** includes a pair of front wheels **102**, a pair of rear wheels **103**, and a chassis **104** (which can also be referred to as a work vehicle frame **104** or tractor frame **104**) coupled to and supported by the wheels **102**, **103**. An operator's cab **105** may be supported by a portion of the chassis **104** and may house various input devices for permitting an operator to control the operation of work vehicle **100** and/or baler **101**. Additionally, work vehicle **100** may include an engine and a transmission mounted on chassis **104**. The transmission may be operably coupled to the engine and may provide variably adjusted gear ratios for transferring engine power to wheels **103** via a drive axle assembly. Control system **114**, in whole or in part, can be coupled with frame **104**.

(9) As shown in FIG. **1**, work vehicle **100** may be coupled to baler **101** via a power take-off (PTO) **106** (which includes a PTO shaft, to which the arrow of **106** points in FIG. **1**) and a tongue **107** to a hitch of work vehicle **100** to allow vehicle **100** to tow baler **101** across the field. As such, work vehicle **100** may, for example, guide baler **101** toward crop material **136** deposited in windrows on the field. As is generally understood, to collect the crop material **136**, baler **101** includes a feeder system **108** (which can be referred to generally as a crop collector) mounted on a front end of baler **101**. Feeder system **108** may, for example, include a pickup assembly **111**, a rotor and augur

arrangement (not shown), and a floor (not shown). Pickup assembly **111** includes a rotating wheel with tines that collect crop material **136** from the ground and direct crop material **136** toward a bale chamber **109** of baler **101** in an overshot manner (rotating clockwise in FIG. **1**), as is known. The rotor and augur arrangement (if used) can be configured to push or otherwise to move crop material **136** towards or into bale chamber **109**, in an undershot manner, as is known. FIG. **1** shows crop material **136** lying in a windrow on the ground of the field and being picked up by pickup assembly **111**. Upon being picked up, crop material **136** flows over pickup assembly **111** and, if used, can flow under the rotor and augur arrangement, to bale chamber **109**. Crop material **136** can move over pickup assembly **111**, as indicated by arrow **132**, which indicates the normal flow direction **132** of crop material **136** relative to feeder system **108**. FIG. **1** shows baler **101** schematically and thus with one pair of wheels, though it can be appreciated that baler **101** can include one or more additional pair of wheels, as in FIG. **2**.
(10) Inside bale chamber **109**, rollers, belts, and/or other devices compact the crop material **136** to form a generally cylindrically-shaped bale **110** for round balers (bale **110** is used for bales that are still forming and bales that are fully formed). Bale **110** is contained within baler **101** until ejection of bale **110** is instructed (e.g., by the operator and/or baler controller **123** of baler **101**). In some embodiments, bale **110** may be automatically ejected from baler **101** once bale **110** is formed, by baler controller **123** detecting that bale **110** is fully formed and outputting an appropriate ejection signal. Further, work vehicle **100** includes control system **114**, which includes controller **115**, which includes a processor **116**, memory **117**, data **118**, and instructions **119**. Control system **114** can further include an input/output device **120** such as a laptop computer (with keyboard and display) or a touchpad (including keypad functionality and a display), device **120** being configured for a user to interface therewith.
(11) As shown in FIG. **1**, baler **101** may also include a tailgate **112** movable between a closed position (as shown in the illustrated embodiment) and an opened position via a suitable actuator assembly. Tailgate **112** and/or the actuator assembly may be controlled to open and close by baler controller **123**. In the closed position, tailgate **112** may confine or retain bale **110** within baler **101**. In the open position, tailgate **112** may rotate out of the way to allow bale **110** to be ejected from the bale chamber **109**. Additionally, as shown in FIG. **1**, baler **101** may include a ramp **113** extending from its aft end that is configured to receive and direct bale **110** away from baler **113** as it is being ejected from bale chamber **109**. In some embodiments, ramp **113** may be spring loaded, such that ramp **113** is urged into a raised position, as illustrated. In such embodiments, the weight of bale **110** on ramp **113** may drive ramp **113** to a lowered position in which ramp **113** directs bale **110** to the soil surface. Once bale **110** is ejected, bale **110** may roll down ramp **113** and be deposited onto the field. As such, ramp **113** may enable bale **110** to maintain its shape and desired density by gently guiding bale **110** onto the field. Further, baler **101** includes control system **122**, which includes controller **123**, which includes a processor **124**, memory **125**, data **126**, and instructions **127**. Controller **123** can communicate with controllers **115**, so that controller **115** outputs information to the display of input/output device **120** of work vehicle **100**, thereby informing a user of various conditions of baler **101** and bales **110** forming or formed therein. Further, baler **101** includes a frame **130** (which can be referred to as baler frame **130**, or more generally as a machine frame **130**) to which all of the components of baler **101** are directly or indirectly coupled. Thus, feeder system **108** and thus also pickup assembly **111** are coupled with frame **130**. Control system **122**, in whole or in part, can be coupled with frame **130**.
(12) It should be appreciated that the configuration of work vehicle **100** described above and shown in FIG. **1** is provided only as one example. Thus, it should be appreciated that the present disclosure may be readily adaptable to any manner of work vehicle configuration. For example, in an alternative embodiment, a separate frame or chassis may be provided to which the engine, transmission, and drive axle assembly are coupled, a configuration common in smaller tractors. Still other configurations may use an articulated chassis to steer work vehicle, or rely on tracks in

lieu of wheels **102**, **103**. Additionally, as indicated previously, work vehicle **100** may, in some embodiments, be configured as an autonomous vehicle. In such embodiments, work vehicle **100** may include suitable components for providing autonomous vehicle operation and, depending on the vehicle configuration, need not include the operator's cab **105**.

(13) Additionally, it should be appreciated that the configuration of baler **101** described above and shown in FIG. **1** is provided only as one example. Thus, it should be appreciated that the present disclosure may be readily adaptable to any manner of baler configuration, or other agricultural machines, such as a vehicle and/or implement, as indicated above. For example, as indicated previously, baler **101** may, in some embodiments, correspond to a square baler configured to generate square or rectangular bales. It should be further appreciated that the illustration of baler **101** in FIG. **1** is schematic.

(14) Further, in general, controllers **115**, **123** may each correspond to any suitable processor-based device(s), such as a computing device or any combination of computing devices. Each controller **115**, **123** may generally include one or more processor(s) **116**, **124** and associated memory **117**, **125** configured to perform a variety of computer-implemented functions (e.g., performing the methods, steps, algorithms, calculations and the like disclosed herein). Thus, each controller **115**, **123** may include a respective processor **116**, **124** therein, as well as associated memory **117**, **125**, data **118**, **126**, and instructions **119**, **127**, each forming at least part of the respective controller **115**, **123**. As used herein, the term “processor” refers not only to integrated circuits referred to in the art as being included in a computer, but also refers to a controller, a microcontroller, a microcomputer, a programmable logic controller (PLC), an application specific integrated circuit, and other programmable circuits. Additionally, the respective memory **117**, **125** may generally include memory element(s) including, but not limited to, computer readable medium (e.g., random access memory (RAM)), computer readable non-volatile medium (e.g., a flash memory), a floppy disk, a compact disc-read only memory (CD-ROM), a magneto-optical disk (MOD), a digital versatile disc (DVD), and/or other suitable memory elements. Such memory **117**, **125** may generally be configured to store information accessible to the processor(s) **116**, **124**, including data **118**, **126** that can be retrieved, manipulated, created, and/or stored by the processor(s) **116**, **124** and the instructions **119**, **127** that can be executed by the processor(s) **116**, **124**. In some embodiments, data **118**, **126** may be stored in one or more databases.

(15) Tractor controller **115**, herein, is assumed to be the primary controller for controlling operations of tractor **100**, and baler controller **123**, herein, is assumed to be the primary controller for controlling operations of baler **101**, though it is understood that at different times each of controllers **115**, **123** can control any of the other of the controllers **115**, **123**. Controllers **115**, **123**, as indicated in FIG. **1**, can be in communication with the other of controllers **115**, **123**, thereby forming unified control system **129**, such that any or all information associated with any controller **115**, **123** can be shared with the other of controllers **115**, **123**, and any controller **115**, **123** can perform the functions of the other controllers **115**, **123**. Controllers **115**, **123** can communicate with each other in any suitable manner, such as a wired connection or a wireless connection, such as radio signals (RF), light signals, cellular, WiFi, Bluetooth, Internet, via cloud-based devices such as servers, and/or the like. Controllers **115**, **123** can be configured to perform any of the functions of any of the other controllers **115**, **123**. Controllers **115**, **123**, **141** can be a part of any network facilitating such communication therebetween, such as a local area network, a metropolitan area network, a wide area network, a neural network, whether wired or wireless. Control system **129**, and controller system **128**, are operatively coupled with tractor **100** and baler **101**, in particular with frames **104**, **130**. According to an embodiment of the present invention, tractor controller **115** can issue commands to baler controller **123**. This is assumed to be the case herein, unless otherwise stated. According to an alternative embodiment of the present invention, baler controller **123** can issue commands to tractor controller **115** (such as for ISOBUS III or higher implements and/or vehicles). This is assumed to be the case herein, unless otherwise stated.

(16) Control system **129** can include additional sensors or other inputs. Control system **114** can further include a GPS (not shown) mounted on tractor **100** (the tractor GPS). The tractor GPS senses the location of tractor **100** within the field, as is known, and this data can be provided to controllers **115**, **123**. Similarly, control system **122** can further include a GPS (not shown) mounted on baler **101** (the baler GPS). The baler GPS senses the location of baler **101** within the field, as is known, and this data can be provided to controllers **115**, **123**. Further, the operator, by way of device **120**, can input or make certain settings. Control system **129** can further include any number additional control systems (with their individual controllers, processors, memory, data, and instructions, substantially similar to what is described above with reference to control systems **114**, **122**), and any such control system can have input/output devices as a part thereof and/or connected thereto.

(17) Work vehicle **100** further includes a ground speed mechanism **134**, coupled with frame **104**. Ground speed mechanism **134** is well-known and thus will not be discussed in detail, but generally serves to cause work vehicle **100** to accelerate, to decelerate, or to maintain a constant speed across the ground, such as a field. For purposes herein, ground speed mechanism **134** can further include a braking system of work vehicle **100**, which is well-known and thus will not be discussed in detail, but generally serves to cause the work vehicle to slow down or to stop. Ground speed mechanism **134** is operatively coupled with control system **114** by, for example, any suitable sensors and actuators known in the art for automatically controlling ground speed mechanism **134** of an automotive vehicle, wherein such sensors and actuators can be deemed to be included within both ground speed mechanism **134** and control system **114**. Such sensors include one or more ground speed sensors **134**A configured for sensing a ground speed of work vehicle **100** and for outputting a ground speed signal corresponding to the ground speed to controller **115**. Such an actuator includes ground speed actuator **134**B configured for receiving a ground speed adjustment signal from controller **115** and thereby for adjusting the ground speed of tractor **100** automatically (and thus also of baler **101**, and thus also of baler frame **130**). The lead line in FIG. **1** for sensor **134**A and actuator **134**B points to a single line, this line being understood to separate the schematic blocks above and below, one such block being sensor **134**A, the other such block being actuator **134**B. Sensor **134**A and actuator **134**B are part of control system **114** and thus also of tractor **100**.

(18) Work vehicle **100**, as indicated above, includes PTO **106**, which includes the PTO shaft (as shown), PTO **106** being coupled with frames **104**, **130**. PTO **106** is well-known and thus will not be discussed in detail, but generally serves to transmit mechanical drive power from an engine of tractor **100** to baler **101** (and thus to various components of baler **101**) via the PTO shaft. PTO **106** can be operatively coupled with control system **114** by, for example, any suitable sensors and actuators known in the art for automatically controlling the torque and speed (revolutions per minute (RPM)) of the PTO shaft, wherein such sensors and actuators can be deemed to be included within both PTO **106** and control system **114**. Such sensors include at least one PTO sensor **106**A configured for sensing a speed of the PTO shaft and for outputting a PTO speed signal corresponding to the speed of the PTO shaft to controller **115**. Such an actuator includes PTO actuator **106**B configured for receiving a PTO speed adjustment signal from controller **115** and thereby for adjusting the speed of the PTO shaft automatically. Actuator **106**B can be, for example, gearing associated with PTO **106** for adjusting the torque and/or speed of the PTO shaft, a transmission of tractor **100**, and/or an actuator associated with an engine of tractor **100** in order to increase or reduce output power of the engine. The lead line in FIG. **1** for sensor **106**A and actuator **106**B points to a single line, this line being understood to separate the schematic blocks left and right, one such block being sensor **106**A, the other such block being actuator **106**B. Sensor **106**A and actuator **106**B as shown in FIG. **1** are part of control system **114** and thus also of tractor **100**. Alternatively or in addition thereto, sensor **106**A and/or actuator **106**B can be located on a baler side of the PTO shaft, and thus communicate directly with controller **123** and thus be a part of control system **122** and baler **101** (as indicated in FIG. **2**).

(19) Referring now to FIG. **2**, there is shown a side view of baler **101** of FIG. **1** in more detail (though still schematically), with portions broken away. Baler **101** further includes top wall **231** of baler **101**, bottom wall **233** of baler **101**, rear wall **237** of baler **101**, side walls **238** of baler **101**, rotary mechanism **239**, belt **240**, belt rolls (which can also be referred to as rollers) **241**, wrapping assembly **242**, and density arm **243** (which can be referred to as device **243**). Bale **110** is essentially fully formed in FIG. **2** and rotates in direction **247** in bale chamber **109**. Top wall **231**, bottom wall **233**, rear wall **237** (which can form part of tailgate **112**), and side walls **238** are coupled with frame **130** and can be formed, for example, of sheet metal, such as steel, or any suitable material. Side walls **235** are lateral walls of baler **101**. Inner surfaces of top wall **231**, bottom wall **233**, rear wall **237**, and side walls **238** can form, at least in part, bale chamber **109**, which is coupled with and/or formed by baler frame **130**. Rotary mechanism **239** is configured to move crop material **136** onward into bale chamber **109** and can rotate clockwise (according to the view of FIG. **2**) about a transversely extending axis of rotation. Belt **240** can be endless and winds in serpentine manner around a series of belt rolls **241**. Belt **240** serves to compress crop material **136** in bale chamber **109** and thereby to form at least a partial circumferential boundary for bale **110** forming in bale chamber **109** (shown extending from about the 9 o'clock to the 6 o'clock positions in FIG. **2**). Belt rolls **241** (only a few of which are labeled in FIG. **2**) serve to tension belt **240**. Roll **241-1** can be deemed to be a drive roll which receives mechanical power from PTO **106** and transmits that mechanical power to belt **240**. Additional rolls (unlabeled) are shown from about the 7 o'clock position of bale **110** to about the 9 o'clock position, these rolls also forming a partial circumferential boundary for bale **110** (and thus compressing bale **110** at least in a latter forming stage of bale **110**), each rotating in a clockwise direction about a respective transversely extending axis of rotation. Wrapping assembly **242** (shown schematically and which is known) is coupled with frame **130** and is configured for wrapping a wrap material (not shown) about crop material **136** in bale chamber **109** during a wrap cycle. Density arm **243** is configured to pivot about axis **243-1** and thus toward or away from belt **240** in order to apply more or less pressure to belt **240**, and thereby to compress bale **110** more or less, and thereby to provide for a more or less dense bale **110**, as is known.
(20) Control system **122** (and thus also baler **101**) includes one or more of the following: belt speed sensor **244**, wheel speed sensor **245-1**A, wheel speed sensor **245-2**A, bale surface speed sensor **246**, drive roll speed sensor **241-1**A, pickup speed sensor **111**A, pickup speed actuator **111**B, density arm position sensor **243**A, and density arm position actuator **243**B. Each sensor **244**, **245-1**A, **245-2**A, **246**, **106**A, **241-1**A, **134**A, **111**A, **243**A (to the extent it is included within a respective embodiment of the present invention) is positioned in any suitable location, is coupled with frame **104**, **130**, and is configured for detecting an operative condition associated with a baling operation and for outputting, to controller **115**, **123**, an operative condition signal corresponding to the operative condition. Controller system **128** is configured for: receiving the operative condition signal; determining at least one operative parameter based at least in part on the operative condition signal(s), the at least one operative parameter being associated with bale **110** of crop material **136** stalling or being about to stall (slowing) in bale chamber **109** of baler **101**. Further, each actuator **134**B, **111**B, **106**B, **243**B (to the extent it is included within a respective embodiment of the present invention) is positioned in any suitable location, is coupled with frame **104**, **130**, and is configured to receive a respective adjustment signal from controller **115**, **123** and thereby to adjust structure respectively associated therewith.
(21) Belt speed sensor **244** can be attached to top wall **231** and positioned within bale chamber **109** so as to be directed at belt **240**. Because belt **240** can be split so as to include at least two circumferential straps (it being assumed herein that belt **240** has two such straps), sensor **244** can be placed off center from a longitudinal midline of baler so as to be directly above a strap of belt **240**. Sensor **244** is configured for detecting a speed of belt **240** and for sending a belt speed signal to controller **115**, **123** corresponding to the belt speed. The speed of belt **240** is associated with a

rotational speed of bale **110** in bale chamber **109**. Optionally, belt speed sensor can be a camera, a lidar sensor, or a radar sensor (or, more generally, a video, optical, or radar sensor).

(22) Wheel speed sensor **245-1**A is configured for detecting a wheel speed of a circumferential wheel **245-1** (that is, wheel mechanism **245-1**) of baler **101**. Wheel **245-1** can be a star-shaped device that includes a plurality of tines that contact crop material **136** on a circumferential surface of bale **110** in bale chamber **109**. Circumferential wheel **245-1** can contact bale **110** between straps of belt **240** and thus be positioned between such straps of belt **240**. As bale **110** rotates, wheel **245-1** is caused to rotate by bale **110**. Wheel speed sensor **245-1**A detects the rotational speed of circumferential wheel **245-1** (which is associated with a rotational speed of bale **110** in bale chamber **109**) and outputs to controller **115**, **123** a circumferential wheel speed signal corresponding to this rotational speed. Wheel **245-1** can be attached to any suitable support coupled with frame **130**, which can move so that wheel **245-1** maintains contact with bale **110** as bale **110** grows from its smallest size to its largest size within bale chamber **109**. Sensor **245-1** can be attached in any suitable way within bale chamber **109**, such as to this support.

(23) Wheel speed sensor **245-2**A is configured for detecting a wheel speed of a side wheel **245-2** (that is, wheel mechanism **245-2**) of baler **101**. Wheel **245-2** can be a star-shaped device that includes a plurality of tines that contact crop material **136** on a lateral end surface of bale **110** in bale chamber **109**. As bale **110** rotates, wheel **245-2** is caused to rotate by bale **110**. Wheel speed sensor **245-2**A detects the rotational speed of side wheel **245-2** (which is associated with a rotational speed of bale **110** in bale chamber **109**) and outputs to controller **115**, **123** a wheel speed signal corresponding to this rotational speed. Wheel **245-2** can be attached to any suitable support coupled with frame **130**, which can move so that wheel **245-2** maintains contact with bale **110** as bale **110** grows from its smallest size to its largest size within bale chamber **109**. Sensor **245-2** can be attached in any suitable way within bale chamber **109**, such as to this support.

(24) Bale surface speed sensor **246** can be attached to top wall **231** and positioned within bale chamber **109** so as to be directed at a circumferential surface of bale **110** between straps of belt **240**. Alternatively, sensor **246** can be placed toward the front of baler, on a respective side wall **238**, rear wall **237**, or bottom wall **233** and thus be directed at a circumferential or a lateral end surface of bale **110**. Sensor **246** is configured for detecting a rotational speed of a surface (i.e., the circumferential surface, or a lateral surface) of bale **110** and for sending a bale surface speed signal to controller **115**, **123** corresponding to the bale surface speed. Optionally, bale surface speed sensor can be a camera, a lidar sensor, or a radar sensor (or, more generally, a video, optical, or radar sensor).

(25) Drive roll speed sensor **241-1**A can be coupled with, for example, a support which supports drive roll **241-1**. Drive roll speed sensor **241-1**A is configured for detecting a rotational speed of drive roll **241-1**A and for sending a drive roll speed signal to controller **115**, **123** corresponding to the rotational speed of drive roll **241-1**. Alternatively or in addition thereto, a sensor can be configured for detecting a rotational speed of any of the other rolls within bale chamber and for sending a roll speed signal to controller **115**, **123** corresponding to the rotational speed of the roll.

(26) Pickup speed sensor **111**A can be a part of pickup assembly **111**. Sensor **111**A can be coupled with any suitable support. Sensor **111**A is configured for detecting a rotational speed of a shaft or reel (hereinafter, the pickup shaft **248**) of pickup assembly **111** which rotates the tines of pickup assembly **111**, and for sending a pickup shaft rotational speed signal to controller **115**, **123** corresponding to the rotational speed of the pickup shaft **248**.

(27) Density arm position sensor **243**A can be a part of density arm **243** and can be coupled with any suitable structure. Sensor **243**A is configured for detecting an angular position of density arm **243** about axis **243-1** and for outputting to controller **115**, **123** a density arm position signal corresponding to the angular position of density arm **243**.

(28) Controller **115**, **123** is configured for receiving the respective signals from sensors **244**, **245-1**A, **245-2**A, **246**, **106**A, **241-1**A, **134**A, **111**A, **243**A. Upon so receiving, controller **115**, **123** is

configured for determining the operative parameter based at last in part on one or more of these signals. More specifically, controller **115**, **123**, can receive a signal from at least one of sensors **244**, **245-1**A, **245-2**A, and **246**, signifying the rotational speed of bale **110**. Further, controller **115**, **123** can receive a signal from at least one of sensors **106**A and **241-1**A, as a reference speed. Controller **115**, **123** can then calculate a difference between, for instance, the rotational speed of belt **240** from sensor **244** and a rotational speed of the PTO shaft (a reference point) from sensor **106**A, and then compare this difference with a predetermined threshold value. If the difference exceeds the predetermined threshold value, then controller **115**, **123** can conclude that bale **109** has stalled within bale chamber **109** (that is, bale **110** has stopped rotating) or is about to stall (bale **110** has slowed substantially), which thus assumes that belt **240** and bale **110** are moving or not moving together. The conclusion of stalled or about to stall is deemed herein to be, collectively, a stall condition (which is an operative parameter). Alternatively or in addition thereto, such calculations of differences can occur relative to any of the sensors **244**, **245-1**A, **245-2**A, **246** (each of which are associated with bale rotational speed) and sensors **106**A, **241-1**A (each of which are associated with a reference point speed). That is, any combination of sensors from the first group including sensors **244**, **245-1**A, **245-2**A, **246** and the second group including sensors **106**A, **241-1**A can be used to make the calculation (at least one from each group being used to calculate the difference). If more than one difference is calculated using a different combination of sensors from these two groups of sensors, then these difference values can be averaged, for instance, to obtain a possibly more accurate reflection of whether a stall condition has occurred. Further, the nature of the stall condition can be further ascertained using these sensors, and a corresponding type of corrective action can be taken by operator and/or controller **115**, **123**. For instance, sensors **245-1**A, **245-2**A, and/or **246** may suggest bale **110** is rotating while sensor **244** suggests that belt **240** has stopped rotating (or vice versa), then a certain type of corrective action can be taken. By way of another example, if each of sensors **244**, **245-1**A, **245-2**A, **246**, and **241-1**A suggest that bale **110**, belt **240**, and drive roll **241-1** are not rotating while PTO sensor **106**A indicates the PTO shaft is rotating, then another type of corrective action can be taken. By way of yet another example, if each of sensors **244**, **245-1**A, **245-2**A, **246**, and **241-1**A suggest that bale **110** and belt **240** are rotating while drive roll **241-1** is not rotating, then another type of corrective action can be taken. By way of yet another example, if sensor **241-1**A indicates drive roll **241-1** is rotating but sensors **244** and sensor **245-1**A (or sensor **245-2**A) indicate that belt **240** and bale **110** are not rotating, then drive roller **241-1** is likely slipping relative to belt **240**, which may require a different type of corrective action (such as increasing the friction by way of density arm **243**); however, the present invention would not be focused on only the scenario of drive roll **241-1** rotating and belt **240** not rotating. Thus, controller **115**, **123** can recognize when the difference (between a value indicate of bale rotation speed (the first group of sensors) and a reference speed (the second group of sensors) is increasing, suggesting that bale rotation speed is dropping (slowing), indicating that stalling is about to occur, or that bale rotation has stopped (stalled). These examples are not intended to be exhaustive.

(29) When controller **115**, **123** finds a stall condition, controller **115**, **123** can output such information (i.e., the various speeds from sensors **244** and **106**A, the difference calculated, and/or a stall condition has occurred (optionally specifying that bale **110** has actually stopped rotating, or is slowing and about to stop) to display **120**, thereby informing the operator, who can then take corrective action to prevent the stall from happening or to undo the stall, such as (for example) decreasing ground speed, decreasing pickup shaft **248** speed, decreasing PTO shaft speed, and/or changing the angular position of density arm **243**, and/or changing these and/or other settings. Alternatively, or in addition thereto, controller **115**, **123** can automatically take corrective action. The corrective action can be directed, for instance, to reducing the quantity of crop material **136** that bale chamber **109** is receiving. Thus, controller **115**, **123** can output an adjustment signal to one or more of actuators **134**B, **111**B, **106**B, **243**B. For instance, controller **115**, **123** can output an

adjustment signal to ground speed actuator **134**B to reduce the speed of tractor **100** and thus also of baler **101**. Alternatively or in addition thereto, controller **115**, **123**, for example, can reduce the speed of rotation of pickup assembly **111** by outputting an adjustment signal to pickup actuator **111**B. Alternatively or in addition thereto, controller **115**, **123**, for example, can reduce the speed of rotation of the PTO shaft by outputting an adjustment signal to PTO actuator **106**B, which can involve changing of gears associated with PTO **106**. Alternatively or in addition thereto, controller **115**, **123**, for example, can change the angular position of density arm **243** by outputting an adjustment signal to density arm actuator **243**B, for instance, to lower the density pressure. Further, pickup sensor **111**A, ground speed sensor **134**A, and density arm position sensor **243**A can be used to know the location of the respective structure sensed by these sensors, so that controller **115**, **123** can know how much to adjust the respective structure. In sum, operative parameters that can be adjusted include at least one of: a ground speed of the baler frame **130**; a speed of the pick-up shaft **248** of pickup assembly **111**; a speed of the PTO shaft of PTO **106**; the angular position of density arm **243**.

(30) Optionally, the bale rotational speed (such as that provided by sensors **244**, **245-1**A, **245-2**A, and/or **246**) can be used in coordination wrapping assembly **242**. That is, the values associated with bale rotational speed can be coordinated with wrap assembly **242**, so as to adjust a wrap tension provided by wrapping assembly **242**, based at least in part on the bale rotational speed that has been monitored. This can beneficial in that it can be determined if the relationship with the wrap material (i.e., net, twine, plastic wrap) speed and the bale rotational speed is changing in a way that might indicate that the wrap material is being stretched and thus has an increased risk of tearing or breaking. Thus, wrapping assembly **242** can include a sensor which senses the wrap speed and outputs a wrap seed signal to controller **123**, which receives the wrap speed signal and compares the values of the bale rotational speed and the wrap speed by calculating a difference therebetween. When the difference exceeds a predetermined threshold, then a stretched, torn, or broken condition of the wrap material can be determined by controller **114**, **123** and outputted to, for example, display **120**, so as to inform operator.

(31) In use, agricultural baler system **135**, including tractor **100** and baler **101**, can include one or more sensors of the first group of sensors **244**, **245-1**A, **245-2**A, **246** and one or more sensors from the second group of sensors **106**A, **241-1**A. Controller **115**, **123** can calculate a difference between a bale rotational speed obtained from a respective sensor from the first group of sensors relative to a reference speed obtained from a respective sensor from the second group of sensors (with any number of combinations of sensors from the two groups) to determine whether bale **110** has stalled or is about to stall in bale chamber **109**. If so, then controller **115**, **123** can output this information to device **120** so as to inform the operator, so that the operator can take corrective action. Alternatively or in addition thereto, controller **115**, **123** can automatically take corrective action, such as by way of outputting adjustment signal(s) to one or more actuators **134**B, **111**B, **106**B, **243**B to make adjustments that can prevent the stall from happening or undoing the stall.

(32) Referring now to FIG. **3**, there is shown a flow diagram showing a method **360** of using an agricultural baler system **135**, the method including the steps of: providing **361** a baler frame **130**, a bale chamber **109** at least one of coupled with and formed by the baler frame **130**, and a control system **129** operatively coupled with the baler frame **130**, the control system **129** including at least one first sensor **244**, **245-1**A, **245-2**A, **246** and a controller system **128** operatively coupled with the at least one first sensor **244**, **245-1**A, **245-2**A, **246**; detecting **362**, by the at least one first sensor **244**, **245-1**A, **245-2**A, **246**, at least one first operative condition associated with a baling operation; outputting **363**, by the at least one first sensor **244**, **245-1**A, **245-2**A, **246**, at least one first operative condition signal corresponding to the at least one first operative condition; receiving **364**, by the controller system **128**, the at least one first operative condition signal; and determining **365**, by the controller system **128**, at least one operative parameter based at least in part on the at least one first operative condition signal, the at least one first operative parameter being associated with

a bale **110** of a crop material **136** one of stalling and being about to stall in a bale chamber **109** of a baler **101**. Further, control system **129** can further include at least one second sensor **106**A, **241-1**A configured for: detecting at least one second operative condition associated with the baling operation; outputting at least one second operative condition signal corresponding to the at least one second operative condition; wherein: the controller system **128** is operatively coupled with the at least one second sensor **106**A, **241-1**A and is configured for: receiving the at least one second operative condition signal; determining the at least one operative parameter based at least in part on the at least one second operative condition signal. Further, the at least one first operative condition can be associated with a rotational speed of the bale **110** in the bale chamber **109** and can be one of: a speed of a belt **240** in the bale chamber **109**; a speed of a wheel mechanism **245-1**, **245-2** in the bale chamber **109**; and a speed of the bale surface of the bale **110**. Further, the at least one second operative condition can be one of: a speed of a power take-off shaft; and a speed of a roll **241-1** of the bale chamber **109**. Further, the at least one operative parameter can be at least one of: a ground speed of the baler frame **130**; a speed of a pick-up **111** of the agricultural baler system **135** which is coupled with the baler frame **130**; a speed of a power take-off shaft **106**; a position of a device **243** of the agricultural baler system **135** which is coupled with the baler frame **130** and is associated with a density of the bale **110**; wherein the controller system **128** is further configured for determining whether one of a stretched condition and a torn condition of a wrap material has occurred.

(33) It is to be understood that the steps of method **360** are performed by controller **115**, **123** upon loading and executing software code or instructions which are tangibly stored on a tangible computer readable medium, such as on a magnetic medium, e.g., a computer hard drive, an optical medium, e.g., an optical disc, solid-state memory, e.g., flash memory, or other storage media known in the art. Thus, any of the functionality performed by controller **115**, **123** described herein, such as the method **360**, is implemented in software code or instructions which are tangibly stored on a tangible computer readable medium. The controller **115**, **123** loads the software code or instructions via a direct interface with the computer readable medium or via a wired and/or wireless network. Upon loading and executing such software code or instructions by controller **115**, **123**, controller **115**, **123** may perform any of the functionality of controller **115**, **123** described herein, including any steps of the method **360**.

(34) The term “software code” or “code” used herein refers to any instructions or set of instructions that influence the operation of a computer or controller. They may exist in a computer-executable form, such as machine code, which is the set of instructions and data directly executed by a computer's central processing unit or by a controller, a human-understandable form, such as source code, which may be compiled in order to be executed by a computer's central processing unit or by a controller, or an intermediate form, such as object code, which is produced by a compiler. As used herein, the term “software code” or “code” also includes any human-understandable computer instructions or set of instructions, e.g., a script, that may be executed on the fly with the aid of an interpreter executed by a computer's central processing unit or by a controller.

(35) These and other advantages of the present invention will be apparent to those skilled in the art from the foregoing specification. Accordingly, it is to be recognized by those skilled in the art that changes or modifications may be made to the above-described embodiments without departing from the broad inventive concepts of the invention. It is to be understood that this invention is not limited to the particular embodiments described herein, but is intended to include all changes and modifications that are within the scope and spirit of the invention.

Claims

1. A control system of an agricultural baler system, the agricultural baler system including a baler frame and a bale chamber at least one of coupled with and formed by the baler frame, the control

system being operatively coupled with the baler frame, the control system comprising: at least one first sensor configured for: detecting at least one first operative condition associated with a baling operation; outputting at least one first operative condition signal corresponding to the at least one first operative condition; a controller system operatively coupled with the at least one first sensor and configured for: receiving the at least one first operative condition signal; determining at least one operative parameter based at least in part on the at least one first operative condition signal, the at least one first operative parameter being associated with a bale of a crop material one of stalling and being about to stall in the bale chamber of a baler, wherein the at least one first operative condition is associated with a rotational speed of the bale in the bale chamber.
2. The control system of claim 1, further including at least one second sensor configured for: detecting at least one second operative condition associated with the baling operation; outputting at least one second operative condition signal corresponding to the at least one second operative condition; wherein: the controller system is operatively coupled with the at least one second sensor and is configured for: receiving the at least one second operative condition signal; determining the at least one operative parameter based at least in part on the at least one second operative condition signal.
3. The control system of claim 2, wherein the at least one operative parameter is at least one of: a ground speed of the baler frame; a speed of a pick-up of the agricultural baler system which is coupled with the baler frame; a speed of a power take-off shaft; a position of a device of the agricultural baler system which is coupled with the baler frame and is associated with a density of the bale.
4. The control system of claim 1, wherein the at least one first operative condition is one of: a speed of a belt in the bale chamber; a speed of a wheel mechanism in the bale chamber; and a speed of the bale surface of the bale.
5. The control system of claim 2, wherein the at least one second operative condition is one of: a speed of a power take-off shaft; and a speed of a roll of the bale chamber.
6. The control system of claim 1, wherein the controller system is further configured for determining whether one of a stretched condition and a torn condition of a wrap material has occurred.
7. An agricultural baler system, comprising: a baler frame; a bale chamber at least one of coupled with and formed by the baler frame; a control system operatively coupled with the baler frame, the control system including: at least one first sensor configured for: detecting at least one first operative condition associated with a baling operation; outputting at least one first operative condition signal corresponding to the at least one first operative condition; a controller system operatively coupled with the at least one first sensor and configured for: receiving the at least one first operative condition signal; determining at least one operative parameter based at least in part on the at least one first operative condition signal, the at least one first operative parameter being associated with a bale of a crop material one of stalling and being about to stall in the bale chamber of a baler, wherein the at least one first operative condition is associated with a rotational speed of the bale in the bale chamber.
8. The agricultural baler system of claim 7, wherein the control system further includes at least one second sensor configured for: detecting at least one second operative condition associated with the baling operation; outputting at least one second operative condition signal corresponding to the at least one second operative condition; wherein: the controller system is operatively coupled with the at least one second sensor and is configured for: receiving the at least one second operative condition signal; determining the at least one operative parameter based at least in part on the at least one second operative condition signal.
9. The agricultural baler system of claim 8, wherein the at least one operative parameter is at least one of: a ground speed of the baler frame; a speed of a pick-up of the agricultural baler system which is coupled with the baler frame; a speed of a power take-off shaft; a position of a device of

the agricultural baler system which is coupled with the baler frame and is associated with a density of the bale.

10. The agricultural baler system of claim 7, wherein the at least one first operative condition is one of: a speed of a belt in the bale chamber; a speed of a wheel mechanism in the bale chamber; and a speed of the bale surface of the bale.

11. The agricultural baler system of claim 8, wherein the at least one second operative condition is one of: a speed of a power take-off shaft; and a speed of a roll of the bale chamber.

12. The agricultural baler system of claim 7, wherein the controller system is further configured for determining whether one of a stretched condition and a torn condition of a wrap material has occurred.

13. A method of using an agricultural baler system, the method comprising the steps of: providing a baler frame, a bale chamber at least one of coupled with and formed by the baler frame, and a control system operatively coupled with the baler frame, the control system including at least one first sensor and a controller system operatively coupled with the at least one first sensor; detecting, by the at least one first sensor, at least one first operative condition associated with a baling operation; outputting, by the at least one first sensor, at least one first operative condition signal corresponding to the at least one first operative condition; receiving, by the controller system, the at least one first operative condition signal; and determining, by the controller system, at least one operative parameter based at least in part on the at least one first operative condition signal, the at least one first operative parameter being associated with a bale of a crop material one of stalling and being about to stall in the bale chamber of a baler, wherein the at least one first operative condition is associated with a rotational speed of the bale in the bale chamber.

14. The method of claim 13, wherein the control system further includes at least one second sensor configured for: detecting at least one second operative condition associated with the baling operation; outputting at least one second operative condition signal corresponding to the at least one second operative condition; wherein: the controller system is operatively coupled with the at least one second sensor and is configured for: receiving the at least one second operative condition signal; determining the at least one operative parameter based at least in part on the at least one second operative condition signal.

15. The method of claim 14, wherein the at least one operative parameter is at least one of: a ground speed of the baler frame; a speed of a pick-up of the agricultural baler system which is coupled with the baler frame; a speed of a power take-off shaft; a position of a device of the agricultural baler system which is coupled with the baler frame and is associated with a density of the bale.

16. The method of claim 13, wherein the at least one first operative condition is one of: a speed of a belt in the bale chamber; a speed of a wheel mechanism in the bale chamber; and a speed of the bale surface of the bale.

17. The method of claim 14, wherein the at least one second operative condition is one of: a speed of a power take-off shaft; and a speed of a roll of the bale chamber.

18. The method of claim 13, wherein the controller system is further configured for determining whether one of a stretched condition and a torn condition of a wrap material has occurred.